

LESSON 3.2

WRITING LARGE NUMBERS USING SCIENTIFIC NOTATION



LESSON INTRODUCTION

MA.8.NSO.1.4 Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

LEARNING TARGETS

- I can express large numbers using scientific notation.
- I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole number exponents and rational number bases.

WORKING TOWARD

MA.912.NSO.1.1 Extend previous understanding of the laws of exponents to include rational number exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

ESSENTIAL QUESTIONS

- How do you write a large number using scientific notation?
- How do you write the standard form of a number expressed in scientific notation?
- How many times larger or smaller is the number expressed in scientific notation than the other number expressed in scientific notation?

LESSON NARRATIVE

Students discover how to write large numbers in scientific notation. They also determine how many times larger or smaller a number written in scientific notation is than another number written in scientific notation. Throughout the lesson students see real-world examples of large numbers and express them in scientific notation.

COMPONENT SUMMARY

3.2.1 WARM-UP (~5 MIN.)

Square a googol

3.2.3 GUIDED INSTRUCTION (10 MIN.)

Write large numbers in scientific notation

3.2.5 COLLABORATION (10 MIN.)

Compare large numbers written in scientific notation

3.2.7 WRAP-UP (~5 MIN.)

Reflect on master of learning targets

3.2.2 EXPLORATION (10 MIN.)

Explore patterns in scientific notation

3.2.4 GUIDED PRACTICE (5 MIN.)

Practice writing large numbers in scientific notation

3.2.6 YOUR TURN! (5 MIN.)

Practice writing and comparing numbers in scientific notation

RESOURCES

GLOSSARY

Key Terms:

- Scientific Notation

Additional Terms:

- N/A

MATERIALS

- (Optional) Highlighters or Colored Pencils

ADDITIONAL PREPARATION

- Consider creating scientific notation templates for students to fill in when expressing numbers in scientific notation.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^5 is two times larger than 10^3 , because $5-3=2$, instead of 10^2 or 100 times larger.

THINGS TO CONSIDER

- Consider creating an anchor chart with the form of scientific notation as $a \times 10^n$ where $1 \leq |a| < 10$. Numbers can be expressed using powers of 10 that are not in this form, but the definition of scientific notation is that the absolute value of a should be at least 1 and less than 10. Consider using both the mathematical representation of the constraint on a as well as the written description of the constraint.
- If time is an issue, then consider assigning 3.2.5 Collaboration and/or 3.2.6 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Turn and Talk

Prompt students to take turns during 3.2.5 Collaboration. Have students to share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

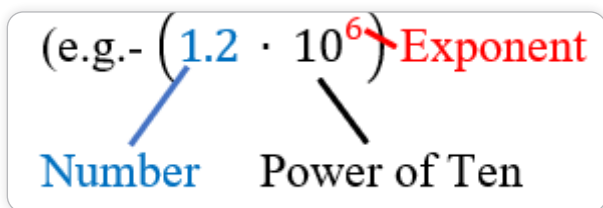
MA.K12.MTR.2.1 Multiple Representations

In 3.2.2 Exploration students explore and discover how to use multiple representations to express large numbers. Engage students in

discussions about how the scientific notation is connected to the standard form of the number.

DIFFERENTIATE INSTRUCTION

In 3.2.3 Guided Instruction, provide a visual entry point by providing students with a template to use when expressing numbers in scientific notation. Students can also create a Venn diagram to record the similarities and differences they notice between the scientific notation and standard form of the values.



3.2.1 WARM-UP: GOOGOL, NOT GOOGLE

TEACHER MOVES

Large numbers could be fascinating for some students and overwhelming to other students. Discuss how expressing the numbers using exponents makes more sense than expressing the numbers in standard form. Prompt students to consider why the popular search engine Google may have chosen a name similar to googol.



3.2.1 Warm-Up: Googol, not Google

A “googol” is a name for a really big number: a 1 followed by 100 zeros.

1. If you square a googol, how many zeros will the answer have? Show your reasoning.

$$(10^{100})^2 = 10^{200}$$

If you square a googol, it will have 200 zeros.



3.2.2 Exploration: Patterns in Scientific Notation

Very large numbers can be written using powers of 10 in a shorthand method called scientific notation.

1. Work with your partner to complete the table.

Power of 10	Value	Scientific Notation	Word Form
10^2	100	1.0×10^2	one hundred
10^3	1,000	1.0×10^3	one thousand
10^4	10,000	1.0×10^4	ten thousand
10^5	100,000	1.0×10^5	hundred thousand
10^6	1,000,000	1.0×10^6	one million
10^9	1,000,000,000	1.0×10^9	one billion
10^{12}	1,000,000,000,000	1.0×10^{12}	one trillion

2. Select two of the numbers. Work with your partner to name something in the world that is described by those numbers.

Answers will vary.

Number	What can be described by this number in the real world?
100	years in a century
1000	millimeters in a meter

3. Work with your partner to describe at least two patterns in the relationships among the power of 10, the numerical value, the scientific notation, and/or the word form.

Answers will vary.

- a. The exponent is the same in the power of ten and in the scientific notation.
- b. The number of zeros in the numerical value is the same as the exponent in the power of ten.

3.2.2 EXPLORATION: PATTERNS IN SCIENTIFIC NOTATION

TEACHER MOVES

Students explore and discover how to use multiple representations to express large numbers. Engage students in discussions about how the scientific notation is connected to the standard form of the number.

DIFFERENTIATE INSTRUCTION

If students finish early, prompt them to list a real-world example for each number in the table. Have a variety of students share their examples so students have the opportunity to see the utility of the lesson learning targets.

3.2.3 GUIDED INSTRUCTION: WRITING LARGE NUMBERS IN SCIENTIFIC NOTATION

TEACHER MOVES

Rewriting numbers in scientific notation should not be taught by moving decimal places, but rather in the context of place value and powers of 10 as built from lesson 1. The Guided Instruction is set up to continue using place value.

DIFFERENTIATE INSTRUCTION

Provide a visual entry point by providing students with a template that uses place value to use when expressing numbers in scientific notation.

TEACHER MOVES

Consider adding the term “Scientific notation” to the classroom Word Wall for students to refer to. Discuss when students have seen scientific notation in their everyday lives and why scientific notation is beneficial.



3.2.3 Guided Instruction: Writing Large Numbers in Scientific Notation

1. Consider the number 2,000,000.
 - a. What power of ten represents 1 million? 10^6
 - b. Complete the multiplication sentence to express 2,000,000 using this power of 10.

$$2 \times 10^6 = 2,000,000$$



Scientific notation is a method of writing very large or very small numbers using exponents in which a number is expressed as the product of a power of 10 and a number that is at least 1 and less than 10.

2. Complete the statement.
When rewriting 2,000,000 in scientific notation, the factor

2 is ☐ less than ☒ greater than or equal to 1 and

☐ less than ☒ greater than or equal to 10^6 . The digit 2 is in the

millions place. A million expressed as a power of 10 is

☐ 10^5 . ☒ 10^6 . Therefore, 2,000,000 is written as ☐ 2×10^5 ☒ 2×10^6

in scientific notation.

3. The expression 20×10^5 is also equivalent to 2,000,000. Use the definition of scientific notation to explain why 20×10^5 is not the correct expression of 2,000,000 in scientific notation.
The factor multiplied by the power of ten must be between 1 and 10.

4. Consider the standard form and scientific notation of the numbers shown in the table.

Standard Form	Scientific Notation
2,300,000	2.3×10^6
2,030,000	2.03×10^6

Describe the similarities and differences between the values expressed in scientific notation.

Similarities	Differences
- Both have a factor being multiplied by 10^6. - Both of the original numbers have a 2 in the millions place. - Both of the original numbers are to the millions. - The factor being multiplied by 10^6 is between 1 and 10.	- The 3 is in a different place in the two numbers in standard form. - The factor being multiplied by 10^6 is not the same.

As explored in the previous lesson, the powers of 10 relate directly to place values.

5. The planet Venus is about 26,000,000 miles from earth.
- What value between 1 and 10 could be multiplied by a power of 10 to result in 26,000,000?
 2.6
 - Which digit is in the ones place of your answer to Part A?
 2
 - In which place value is that digit in 26,000,000?
 ten million
 - Which power of 10 relates to this place value?
 10^7
 - Express 26,000,000 in scientific notation.
 2.6×10^7
6. In 2018, funding for education in Florida totaled \$30,100,000,000. Express this value in scientific notation.
 3.01×10^{10}

3.2.3 GUIDED INSTRUCTION: WRITING LARGE NUMBERS IN SCIENTIFIC NOTATION (CONTINUED)

TEACHER MOVES

Spend time discussing the difference between the two values of a in question 4. Focus students' attention to the zero in 2.03 and that there is no zero in 2.3. Point students to the anchor chart that shows that the number, a , in $a \times 10^n$ must fit the constraint $1 \leq |a| \leq 10$ to be considered scientific notation. Numbers written using powers of 10 where the number a does not fit this constraint are, by definition, not considered to be written in scientific notation.

DIFFERENTIATE INSTRUCTION

Students can create a Venn diagram to record the similarities and differences they notice between the scientific notation and standard form of the values.

TEACHER MOVES

Students may struggle determining the value of a . Refer students back to question 4 for an example.

3.2.4 GUIDED PRACTICE: WRITING LARGE NUMBERS IN SCIENTIFIC NOTATION

TEACHER MOVES

The connection to theme parks will likely ignite student interest. When debriefing the activity, call on a variety of students to share their answers and thinking. Then, poll the class to see if they agree or disagree and why. Ask each student who shares if they have visited the theme park or which theme park is their favorite.



3.2.4 Guided Practice: Writing Large Numbers in Scientific Notation

- The table shows the number of visitors to several amusement parks in Florida in 2019. Express each value in scientific notation.

Florida Theme Park	Approximate Number of Visitors in 2019	Equivalent Value in Scientific Notation
Magic Kingdom	20,960,000	2.096×10^7
Animal Kingdom	13,900,000	1.39×10^7
Busch Gardens	4,200,000	4.2×10^6
Sea World	4,640,000	4.64×10^6
Adventure Island	656,000	6.56×10^5



3.2.5 Collaboration: Comparing Large Numbers Written in Scientific Notation

1. Work with your partner to complete the following.

- a. Complete the table.

Scientific Notation	Standard Form
2×10^2	200
2×10^3	2,000
2×10^4	20,000
2×10^5	200,000

- b. Complete each statement.

2×10^3 is ☐ 10 ☐ 100 ☐ 1000 times ☐ smaller ☒ larger than 2×10^2 .

2×10^2 is ☐ 10 ☒ 100 ☐ 1000 times ☒ smaller ☐ larger than 2×10^4 .

2×10^5 is ☐ 10 ☐ 100 ☒ 1000 times ☐ smaller ☒ larger than 2×10^2 .

- c. Describe how to compare numbers written in scientific notation if the factors being multiplied by a power of 10 are the same.

You can compare the powers of ten.

2. Complete the table.

Scientific Notation	Standard Form
3.2×10^7	32,000,000
6.4×10^7	64,000,000
9.6×10^7	96,000,000

- a. How many times larger is 6.4×10^7 than 3.2×10^7 ?

It is two times larger.

- b. How many times smaller is 3.2×10^7 than 9.6×10^7 ?

It is three times smaller.

- c. Describe how to compare numbers written in scientific notation if the powers of 10 are the same.

You should compare the factors being multiplied by the power of ten.

3. Complete the table.

Scientific Notation	Standard Form
2.1×10^4	21,000
4.2×10^5	420,000
8.4×10^6	8,400,000

- a. How many times larger is 4.2 than 2.1?

- b. How many times larger is 10^5 than 10^4 ?

- c. How many times larger is 4.2×10^5 than 2.1×10^4 ?

= times larger

- d. Complete the statement.

The value of 8.4×10^6 is ☐ 4 ☐ 40 ☒ 400 times larger than

the value of 2.1×10^4 because 8.4 is ☐ 4 ☐ 40 ☐ 400 times

larger than 2.1×10^4 is ☐ 10 ☒ 10^2 ☐ 10^3 times larger than

10^4 , and = ☐ 4. ☐ 40. ☒ 400.

If the factors being multiplied by the powers of 10 are different, but one is not a multiple of the other, the comparison of the two values can be *approximated*.

3.2.5 COLLABORATION: COMPARING LARGE NUMBERS WRITTEN IN SCIENTIFIC NOTATION

TEACHER MOVES

Students may incorrectly think that the difference in the powers of 10 is how many times larger or smaller a number is than another number. For example, they may think 10^5 is two times larger than 10^3 , because $5 - 3 = 2$. Remind students that it is 10^2 or 100 times larger.

TEACHER MOVES

Take Turns

Prompt students to take turns during 3.2.5 Collaboration. Have students to share their strategies and answers. Then, ask the partner to share if they agree or disagree and why.

3.2.5 COLLABORATION: COMPARING LARGE NUMBERS WRITTEN IN SCIENTIFIC NOTATION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider reading the text-dense problems aloud in this and other components to provide an auditory entry point.

TEACHER MOVES

Consider having students multiply 400 times 2.38855×10^5 to verify their answer.

4. The average distance from Earth to the moon is 2.38855×10^5 miles. The average distance from Earth to the sun is about 9.3×10^7 . Complete the statements to approximate how many times farther the sun is from Earth than the moon.

9.3 is about

☐ 2
☒ 4
☐ 6

 times

☐ smaller
☒ larger

 than 2.38855×10^5 and

10^7 is

☐ 10
☒ 10^2
☐ 10^3

 times

☐ smaller
☒ larger

 than 10^5 , so the value of 9.3×10^7 is about 400 times larger than 2.38855×10^5 .



3.2.6 Your Turn!

- Each statement contains a quantity. Rewrite each quantity in scientific notation.
 - There are about 37,000,000,000 cells in the average human body.
 3.7×10^{10}
 - Light travels through space at a rate of 300,000,000 meters per second.
 3×10^8
- The population of Florida was about 2.22×10^7 in 2021. The population of the world was about 7.87×10^9 in the same year.

Complete the statement.

In 2021, the population of the world was about 400 times larger than the population of Florida.

3.2.6 YOUR TURN!

TEACHER MOVES

Remind students that the word ‘about’ in question 2 directions indicates that they can estimate how many times larger the population of the world is than the population of Florida.



3.2.7 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is related to each target.

- a. I can express large numbers using scientific notation.

Answers will vary.

I need help.

I'm getting there, but

I understand this

I can't get started.

I need more practice.

and I feel confident.



- b. I can determine how many times larger or smaller a number is compared to another expressed in scientific notation.

Answers will vary.

I need help.

I'm getting there, but

I understand this

I can't get started.

I need more practice.

and I feel confident.



3.2.7 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Use data from this formative assessment to determine which students need remediation before the next lesson.